PRODUCT SERIES

3

Terminal Equipment

INTEGRATED CARDPHONE SYSTEM



INTRACOM

HELLENIC TELECOMMUNICATIONS AND ELECTRONICS INDUSTRY

PRODUCT SERIES

3

Terminal Equipment

INTEGRATED CARDPHONE SYSTEM

The on-going development of new telecommunication means enables telecom operators to provide their customers with higher quality of service.

In the public telephony, the traditional coin operated phones were always considered as a service to the public, rather than profitable business. Vandalism, fraud, low quality of service and high operational costs were always disadvantages for such systems.

A modern payphone network operator has to provide a customer focused, demand driven service, built on strong technical marketing and commercial foundations. Such a demand can be satisfied with the modern card operated payphones.

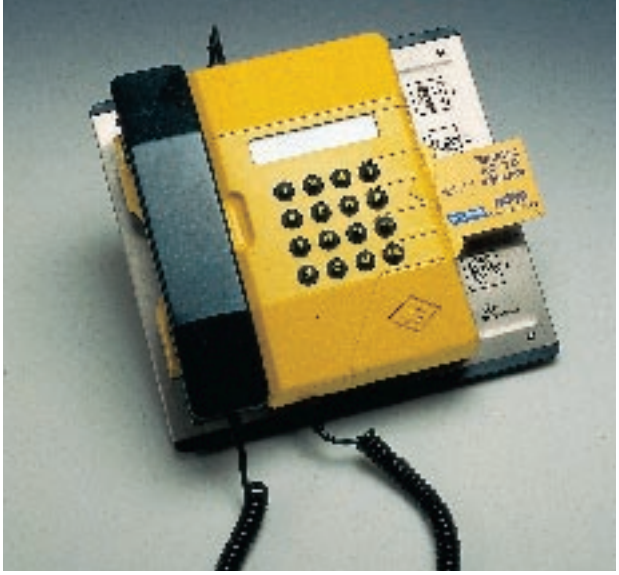
Telecom operators, worldwide, invest highly in advanced cardphones, remotely supervised by management systems, while replacing the traditional coin operated public phones.

The operation of a cardphone system has been proven to be, not only a high quality service to the public, but also a direct source of revenue for the operator.

INTRACOM offers an integrated cardphone system consisting of modern public terminals for both indoor and outdoor use, supervised by a centralised network management system.

INTRACOM's integrated cardphone system provides cost-effective solutions to a variety of market applications.





System Components

INTRACOM's cardphone system comprises the following components:

INT 200

Wall mounted, all weather, outdoor type cardphone, with high mechanical strength and resistance to vandalism.

INT 100

Table top or wall mounted cardphone, for supervised indoor premises.
Ideal for places where a heavy duty cardphone is not suitable.

CSMS

Cardphone Supervision & Maintenance System.

Line Unit

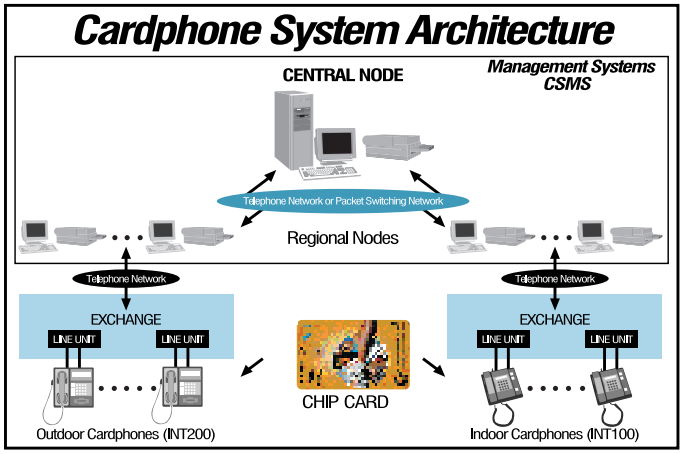
Debiting and line protection unit against fraudulent use.

System Operation

INTRACOM's cardphone system operates under an off-line scheme. Each time a telephone card is inserted to a cardphone, it is checked for its validity. After a successful call, the data regarding this transaction are recorded in the cardphone's non-volatile memory. Each cardphone reports to the CSMS at a predetermined time, via the PSTN, through a built- in modem, in order to update the system database.
Each update call is twofold. First the cardphone reports to the CSMS all transaction and operational data. Second the CSMS updates the cardphone on its parameters (operational parameters, new card series, black lists, etc).
In addition, the cardphones can initiate an alarm call to the CSMS immediately after an urgent event (e.g. a handset removal or a card reader failure) is detected. In special cases the cardphone can make an update call after a "call me back" call from the CSMS. The Line Unit is interposed between the exchange and the cardphone unit. It communicates constantly with the cardphone through a specific handshake protocol.
Additionally, it decodes the debiting criterion arriving from the exchange, whether it is a DC or a 12/16 KHz pulse and transmits it, as a debiting pulse, towards the cardphone. Failure of the cardphone to respond correctly, causes the line unit to release the line. Thus, in case an ordinary telephone is connected in parallel with the cardphone line, the Line Unit detects the fraud attempt and releases the call.

Cardphone Supervision & Maintenance System

INTRACOM's Cardphone System features an easy to use, graphical windows based Management System for Administration and Maintenance of Cardphones as well as for Card and Transaction Management.
All types of Cardphone Terminals offered by INTRACOM are supported by the same Management System, which takes into account all individual requirements of each type of Terminal.



Cardphone Features

CARD	Prepaid telephone card, chip card technology. Eurochip I & II, T2G, 104 bit EEPROM, 256 bit EPROM. Five sockets for Security Modules (SAM). Subscriber card. Optionally magnetic credit card. Technician card for maintenance personnel.
MAINTENANCE	Modular structure with common spare parts for both, INT100 and INT200. Parametrical operation. Self test capability. Supervision from remotely placed Management Systems (CSMS). Alarm calls in case of serious malfunction.
CONNECTION TO THE EXCHANGE	Through the a and b line pair only. No batteries or external power supply required.
TYPES OF CALLS	Local, long distance, international calls. Free of charge calls to emergency numbers, without card. Blocked numbers.
DEBITING	16 KHz / 12 KHz pulses. Self generated debiting capability. Press to talk button option.
ADDITIONAL FEATURES	Complete S/W downloading, service log, backlit display, activated display during on-hook.
SECURITY	Eurochip Active authentication. «White» and «black» list of telephone cards. Transactions monitoring.
USER INTERFACE	User friendly. Alphanumeric or Graphics display. Multi-Language support. Card change, follow on call, volume control, redial key. Hearing aid coupling.
COMPATIBILITY	Same main board and card-reader for both, INT100 and INT200.
ACCESSORIES OF INT 100	• Inclined plastic base for permanent fixing horizontally or vertically. • Terminal adapter for fax, modem etc.
OPTION FOR INT 200	• GSM Interface
LINE UNIT	Detection of debiting criteria received from the exchange (DC, 12/16 KHz), generation of debiting pulses (12/16 KHz) towards the cardphone and release of the call in case an ordinary phone is connected to the payphone line (Antifraud function).

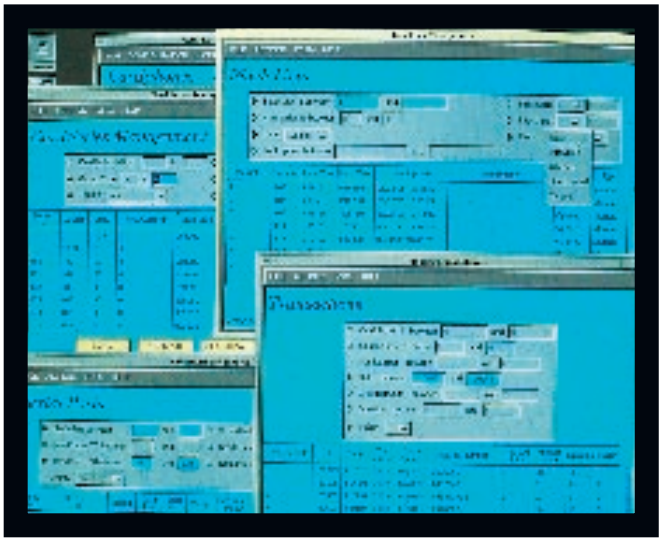
CSMS Regional Node

The CSMS regional node is responsible for Technical Maintenance and Administration of all the cardphones in a region. The Technical Maintenance part receives all cardphone calls that can either be routine reporting of data, or alarm reporting that requires immediate attention and action by the maintenance personnel. During a routine call, all operational parameters that need to be changed for each individual cardphone are also downloaded.

CSMS Central Node

The CSMS central node is responsible for Network Management of the regionally operated nodes and includes central administration, network configuration and generation of total or regional statistics on various aspects

Integrated Cardphone System

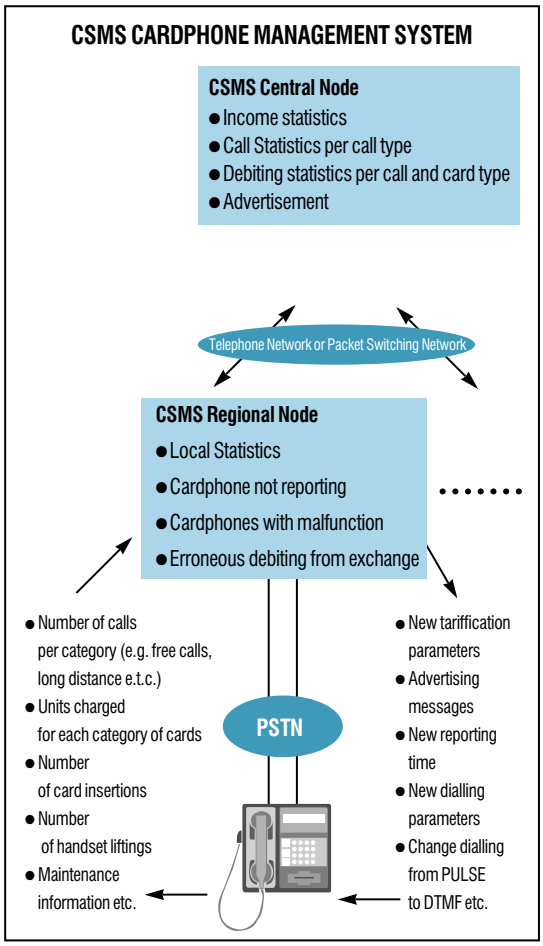


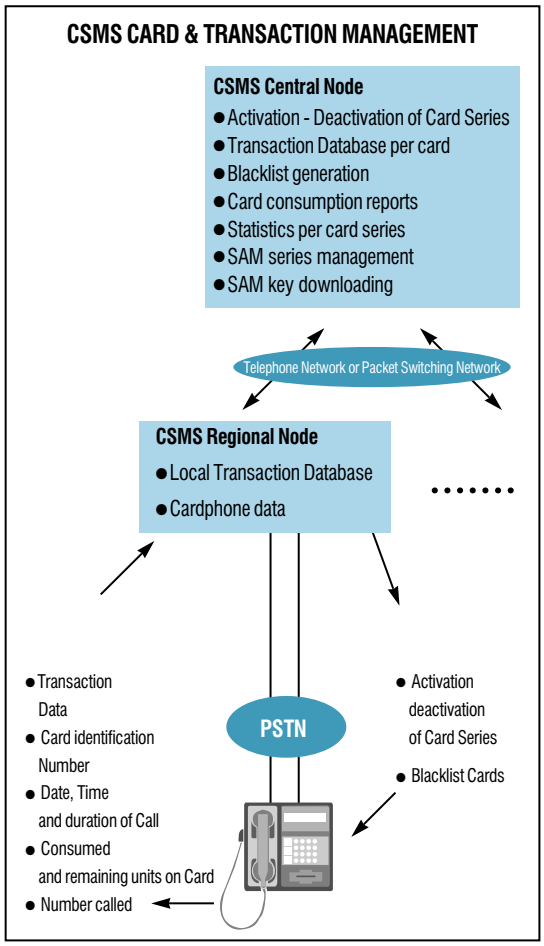
of the Cardphone's operation and usage.

Card & Transaction Management for Central and Regional Nodes

This part of the system is responsible for Card and Transaction Management, as well as for high level security against fraudulent usage of Cards. The Card Management part is responsible for activating Card Series before circulation, or deactivating them, if needed.

The Transaction Management part is responsible for processing and compiling of transactions. Under certain conditions a card can be blacklisted either automatically, in case the system detects fraudulent transactions, or manually. Various statistical reports related to either Cardphone or Card usage, can be generated, including consumption, average consumption etc.





Technical Specifications

INT 100/INT 200

TYPE OF CARD Chip card technology according to ISO standards. Eurochip I & II, T2G, 104-bit EEPROM, 256-bit EPROM, µ Proccessor Cards. Optional, magnetic credit card, concurrently with chip card. Technician card.

DISPLAY STN technology. Alphanumeric 2 lines x 20 characters or Graphics display . Advertising messages capability, downloaded from the CSMS. Optional backlit display.

MODEM V.23 asynchronous built-in modem.

KEYPAD Twelve (12) standard keys for dialing (DTMF or pulse) according to ITU-T recommendations. Four functions keys (software programmable) are available in order to perform any of the following operations:

- push - to - talk
- language change
- card change
- follow-on call
- redial
- volume control

HANDSET Dynamic capsules. Hearing aid coupling according to ITU-T recommendations.

CARD READER Modular card reader for optimal maintenance and replacement. For INT 200, protection against insertion of foreign objects.

RINGER Incoming signals are indicated by a two tone ringer mounted inside the cardphone.

ANTI FRAUD SYSTEM The operation of the cardphone, in conjunction with an additional Line Unit placed in the exchange, provides a high degree of security against fraudulent use of the line pair (another telephone set alone, or in parallel with the cardphone).

LINE PROTECTION Overvoltage and EMI protection according to IEC standards

POWER Line powered, ranging from 18 mA to 100 mA.

MATERIAL INT 100: Thermoplastic ABS Cycolac™.
INT 200 Door : Aluminium alloy
Box: Mild steel sheet special treated against corrosion. Electrostatically painted with polyester type powder paint.

DIMENSIONS INT 100: H 75 mm x W 189 mm x D 220 mm with handset (without inclined base).
INT 200: H 275 mm x W 275 mm x D 172 mm (with handset)

WEIGHT INT 100: 1 Kg (3 Kg with inclined base).
INT 200: 6.5 Kg.

ENVIRONMENTAL INT 100: 0°C to +50°C.
up to 80% RH at 35°C.
INT 200: -30°C to +60°C.
up to 95% RH at 40°C.

Specifications subject to change without prior notice

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